Understanding Cyclists' Visual Behavior Using Eye-Tracking Technology: A Systematic Review

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PMC PDF Link: https://pmc.ncbi.nlm.nih.gov/articles/PMC11723293/pdf/

DOI Link: https://doi.org/10.3390/s25010022

Abstract

Eye-tracking technologies are emerging in research aiming to understand the visual behavior of cyclists to improve their safety. These technologies gather real-time information to reveal what the cyclists look at and how they respond at a specific location and time. This systematic review investigates the use of eye-tracking systems to improve cyclist safety. An extensive search of the SCOPUS and WoS databases, following the PRISMA 2020 guidelines, found 610 studies published between 2010 and 2024. After filtering these studies according to predefined inclusion and exclusion criteria, 25 were selected for final review. The included studies were conducted in real traffic or virtual environments aiming to assess visual attention, workload, or hazard perception. Studies focusing on other types of road users or participants not involved in active cycling were excluded. Results reveal the important impact of road elements' design, traffic density, and weather conditions on cyclists' gaze patterns. Significant visual workload is imposed mainly by intersections. Along with the valuable insights into cyclist safety, potential biases related to small sample sizes and technological limitations were identified. Recommendations for future research are discussed to address these challenges through more diverse samples, advanced technologies, and a greater focus on peripheral vision.